

Measurement of π^0 and η mesons in oxygen-oxygen collisions at
 $\sqrt{s_{\text{NN}}} = 5.36$ TeV using Machine Learning-based photon
identification in ALICE Run 3

Supplemental figures

Machine-learning hyperparameter optimisation in oxygen–oxygen collisions
at $\sqrt{s_{\text{NN}}} = 5.36$ TeV

by Charlize Gohmann

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1 Hyperparameter optimisation history

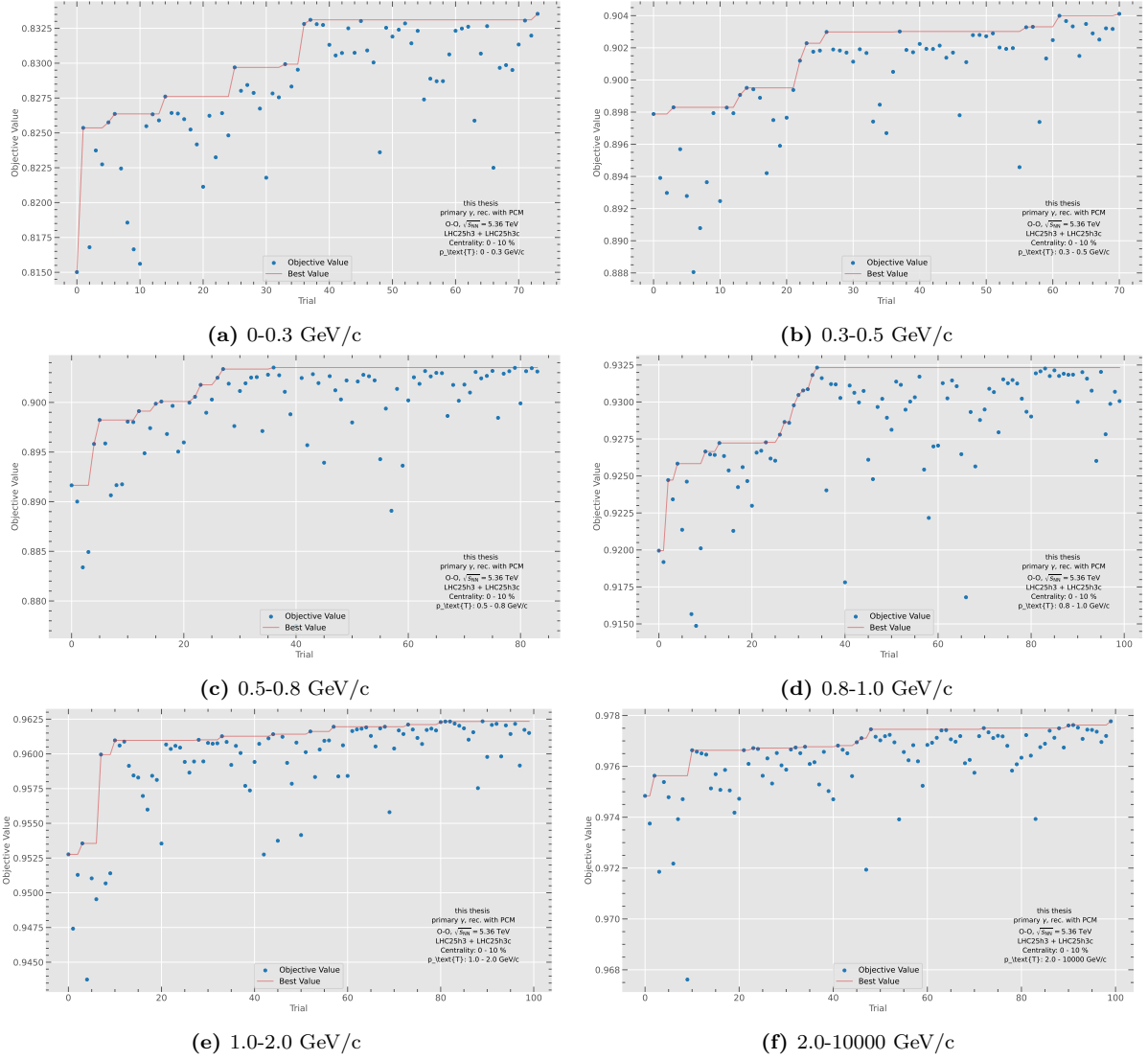
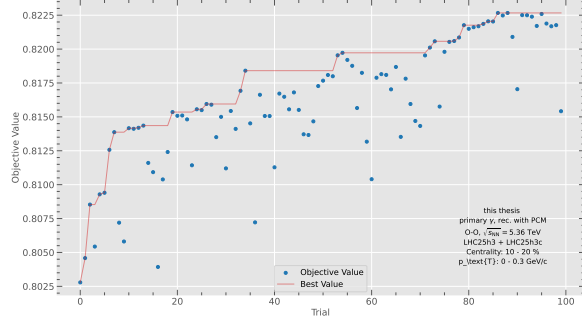
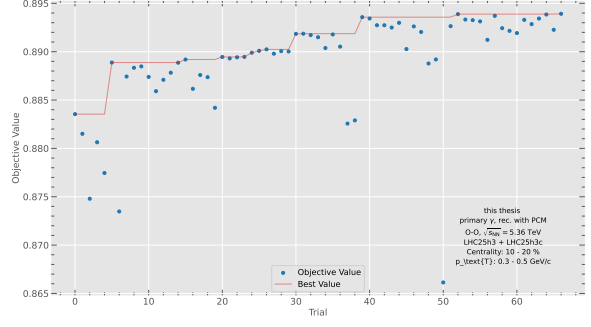


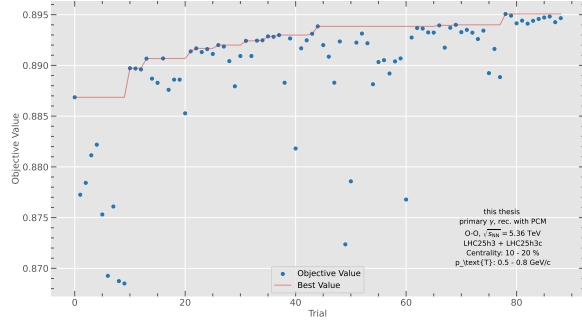
Figure 1: Hyperparameter optimization history for all p_T bins in the centrality range 0-10 %



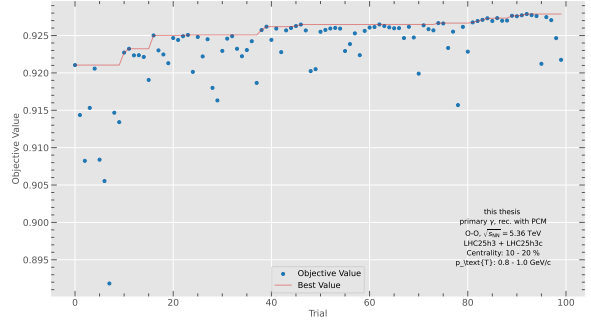
(a) 0-0.3 GeV/c



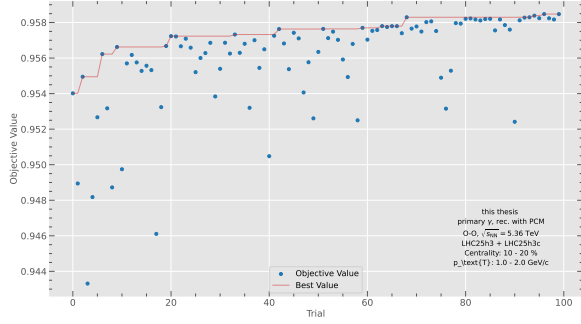
(b) 0.3-0.5 GeV/c



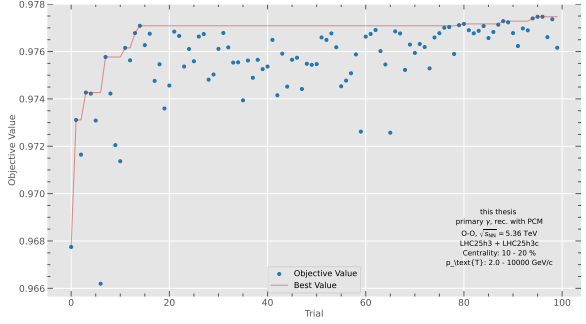
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

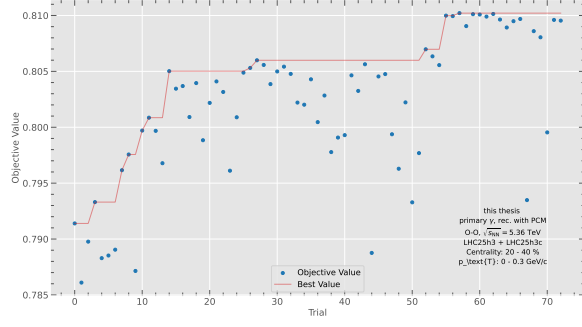


(e) 1.0-2.0 GeV/c

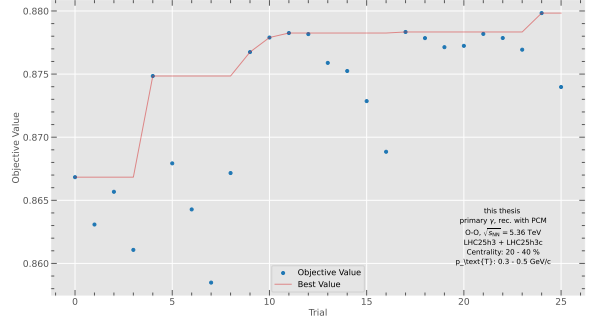


(f) 2.0-10000 GeV/c

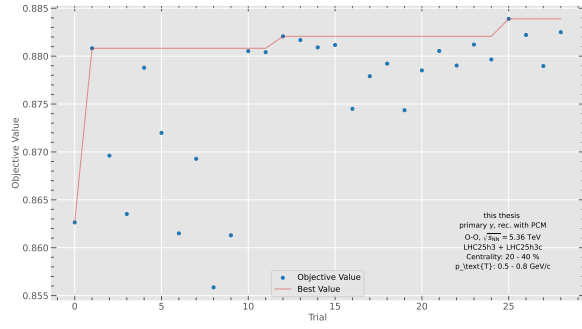
Figure 2: Hyperparameter optimization history for all p_T bins in the centrality range 10-20 %



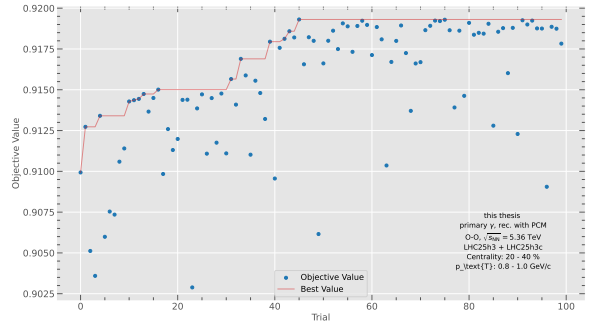
(a) 0-0.3 GeV/c



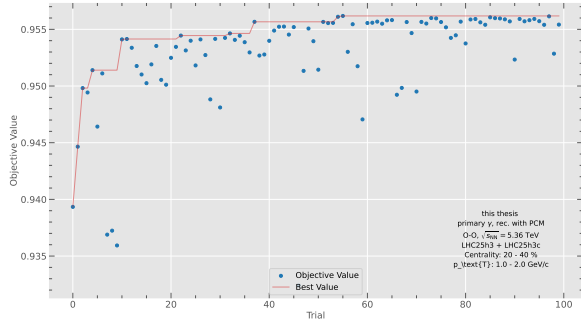
(b) 0.3-0.5 GeV/c



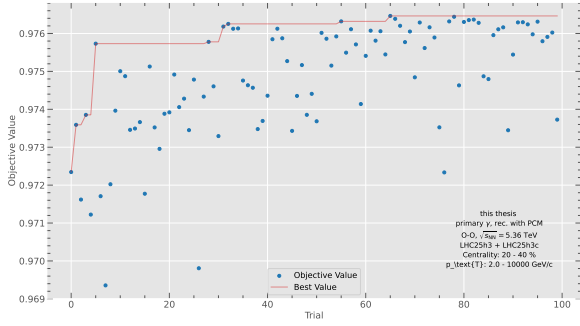
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

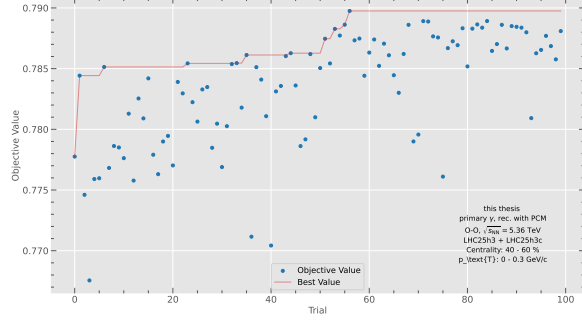


(e) 1.0-2.0 GeV/c

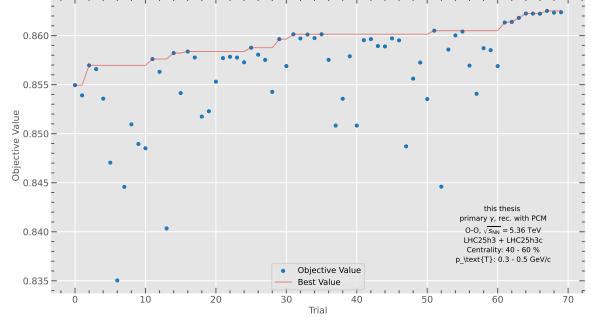


(f) 2.0-10000 GeV/c

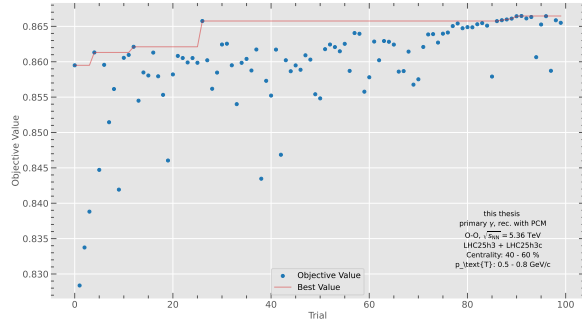
Figure 3: Hyperparameter optimization history for all p_T bins in the centrality range 20-40 %



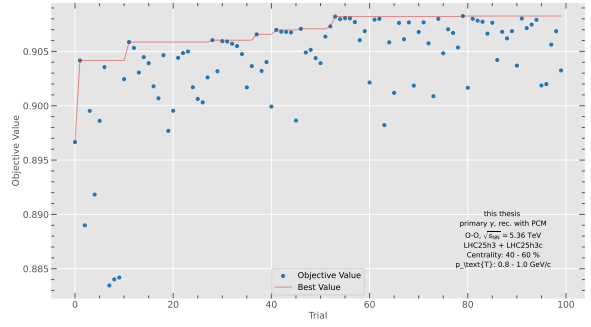
(a) 0-0.3 GeV/c



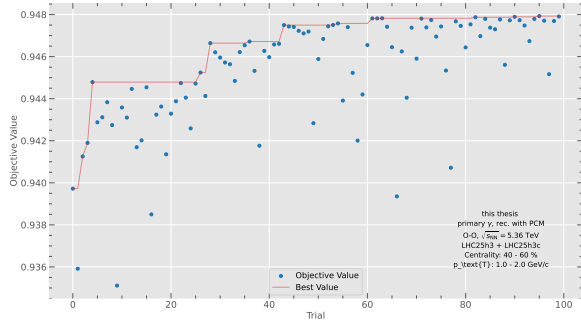
(b) 0.3-0.5 GeV/c



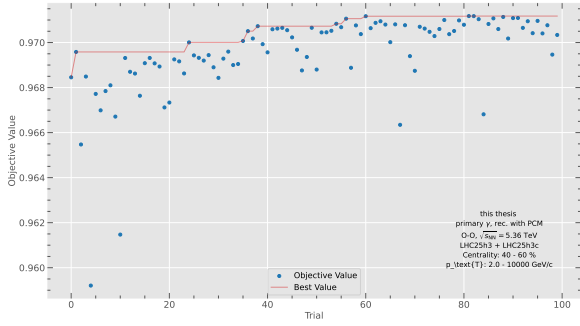
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

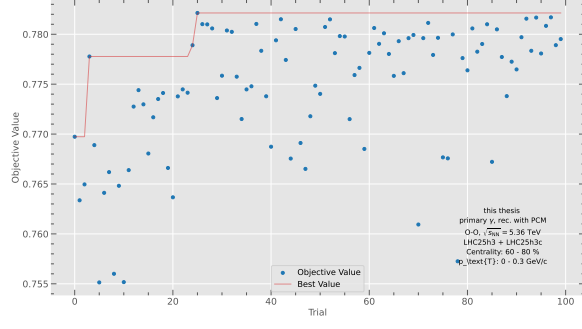


(e) 1.0-2.0 GeV/c

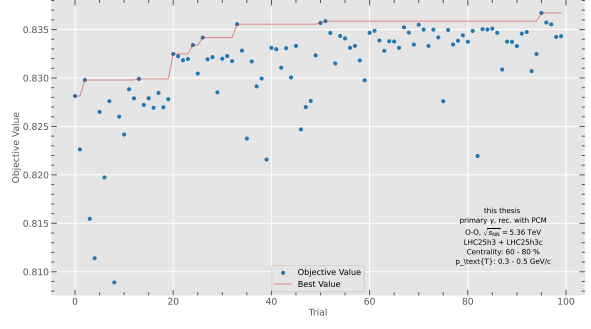


(f) 2.0-10000 GeV/c

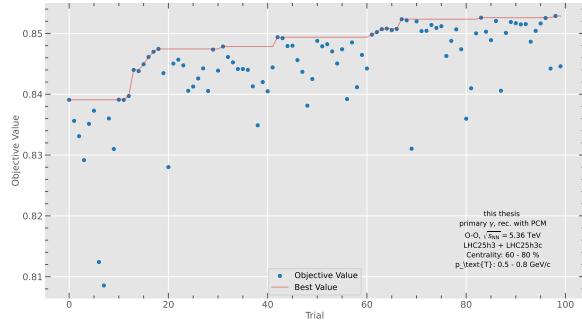
Figure 4: Hyperparameter optimization history for all p_T bins in the centrality range 40-60 %



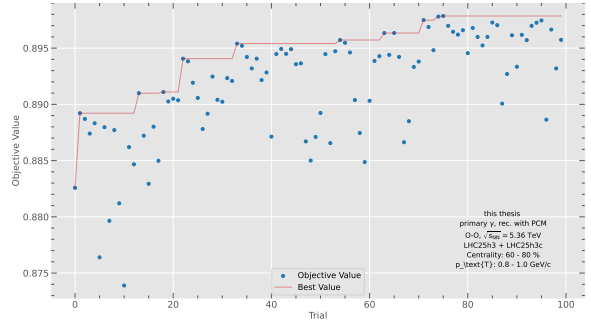
(a) 0-0.3 GeV/c



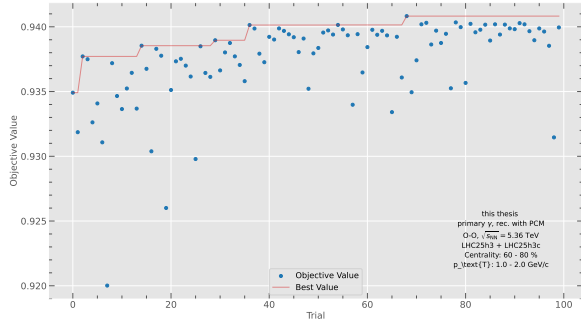
(b) 0.3-0.5 GeV/c



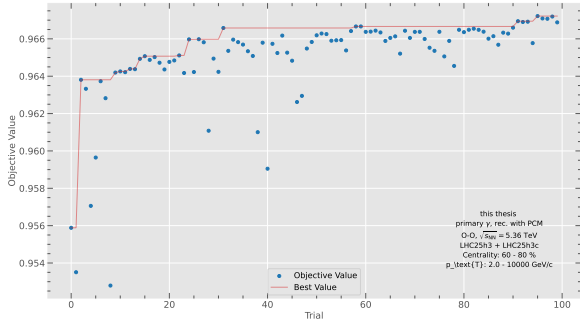
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

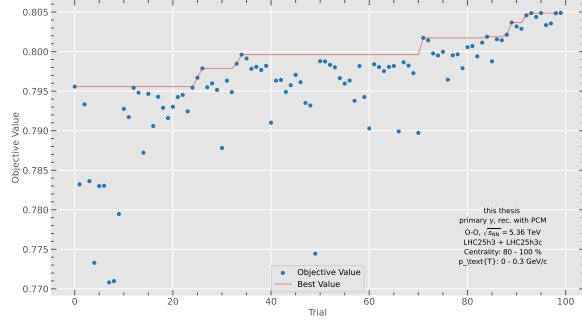


(e) 1.0-2.0 GeV/c

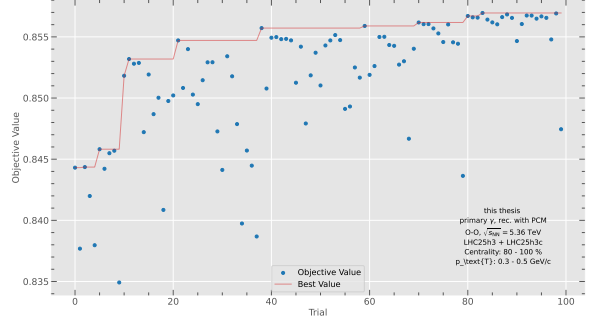


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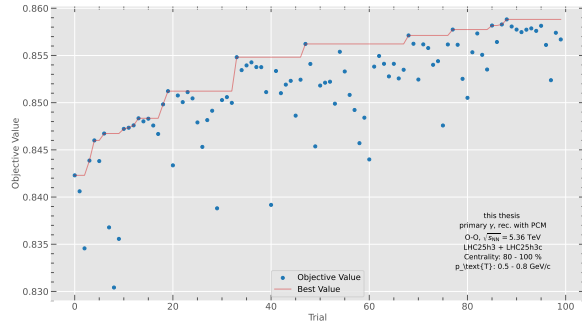
Figure 5: Hyperparameter optimization history for all p_T bins in the centrality range 60-80 %



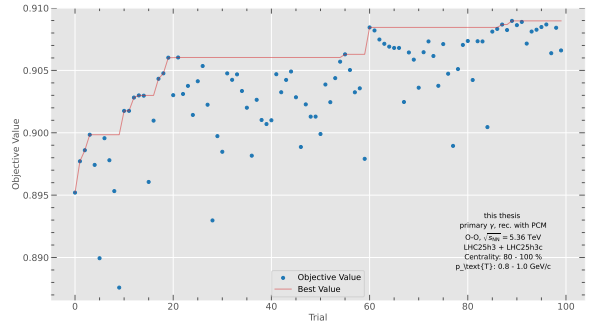
(a) 0-0.3 GeV/c



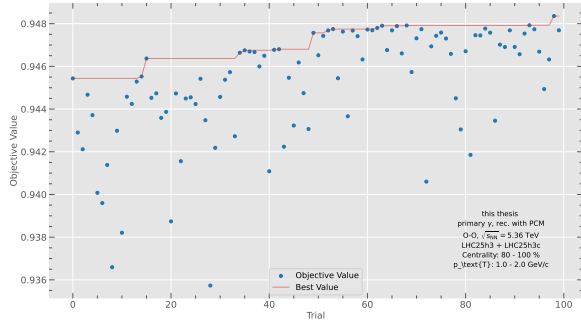
(b) 0.3-0.5 GeV/c



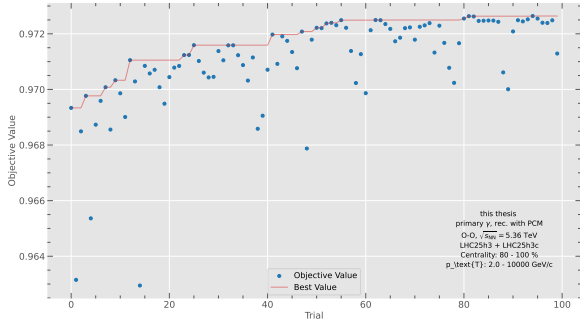
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c



(e) 1.0-2.0 GeV/c



(f) 2.0-10000 GeV/c

Figure 6: Hyperparameter optimization history for all p_T bins in the centrality range 80-100 %

2 Hyperparameter evolution throughout the tuning process

2.1 Gamma, learning rate, maximum depth, minimum child weight, and number of estimators



Figure 7: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 0-10 %



Figure 8: Hyperparameter evolution of γ , learning_rate , max_depth , min_child_weight , and n_estimators throughout the tuning process for all p_T bins in the centrality range 10-20 %

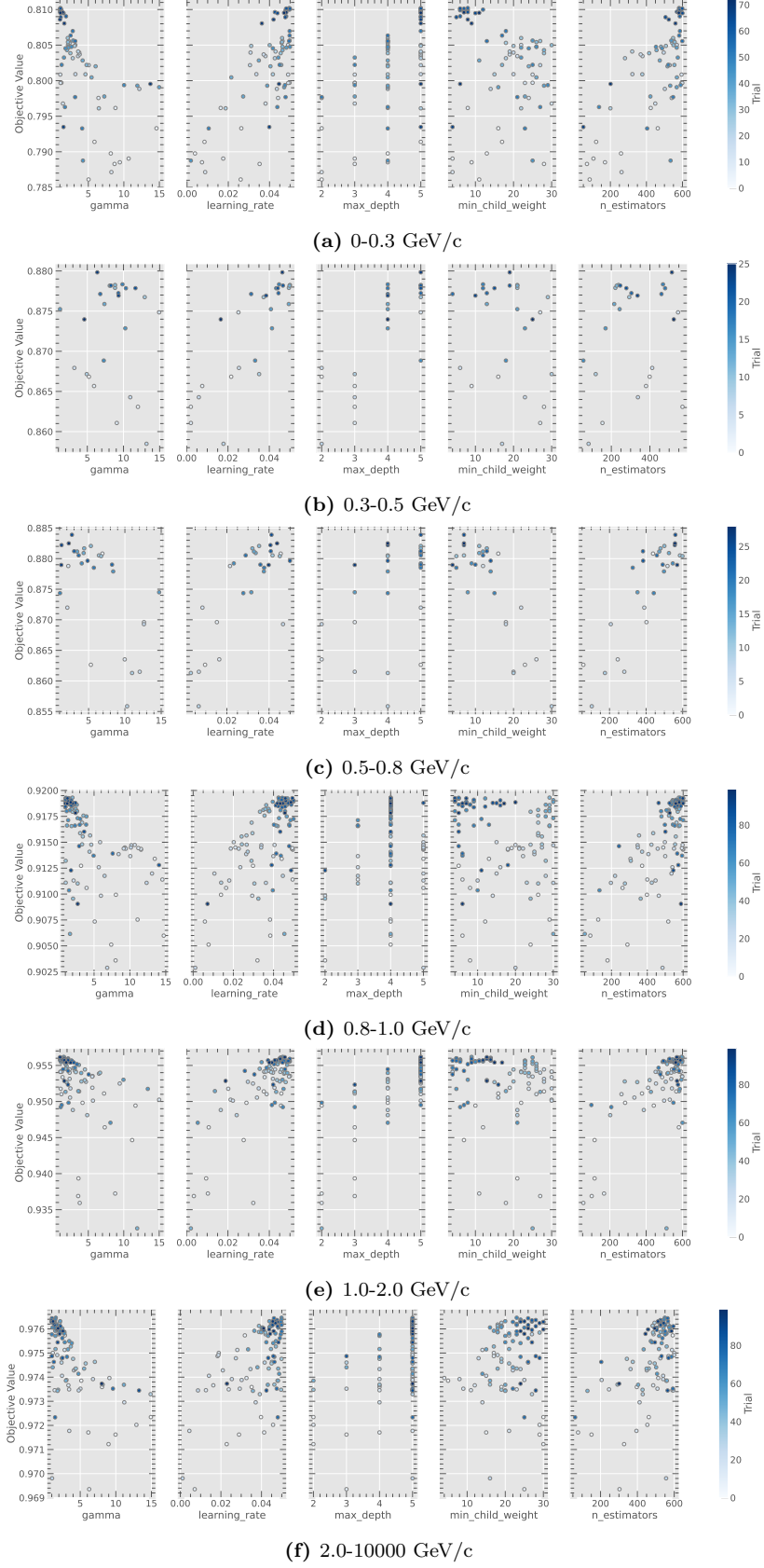


Figure 9: Hyperparameter evolution of γ , learning_rate , max_depth , min_child_weight , and n_estimators throughout the tuning process for all p_T bins in the centrality range 20-40 %



Figure 10: Hyperparameter evolution of γ , learning_rate, max_depth, min_child_weight, and n_estimators throughout the tuning process for all p_T bins in the centrality range 40-60 %

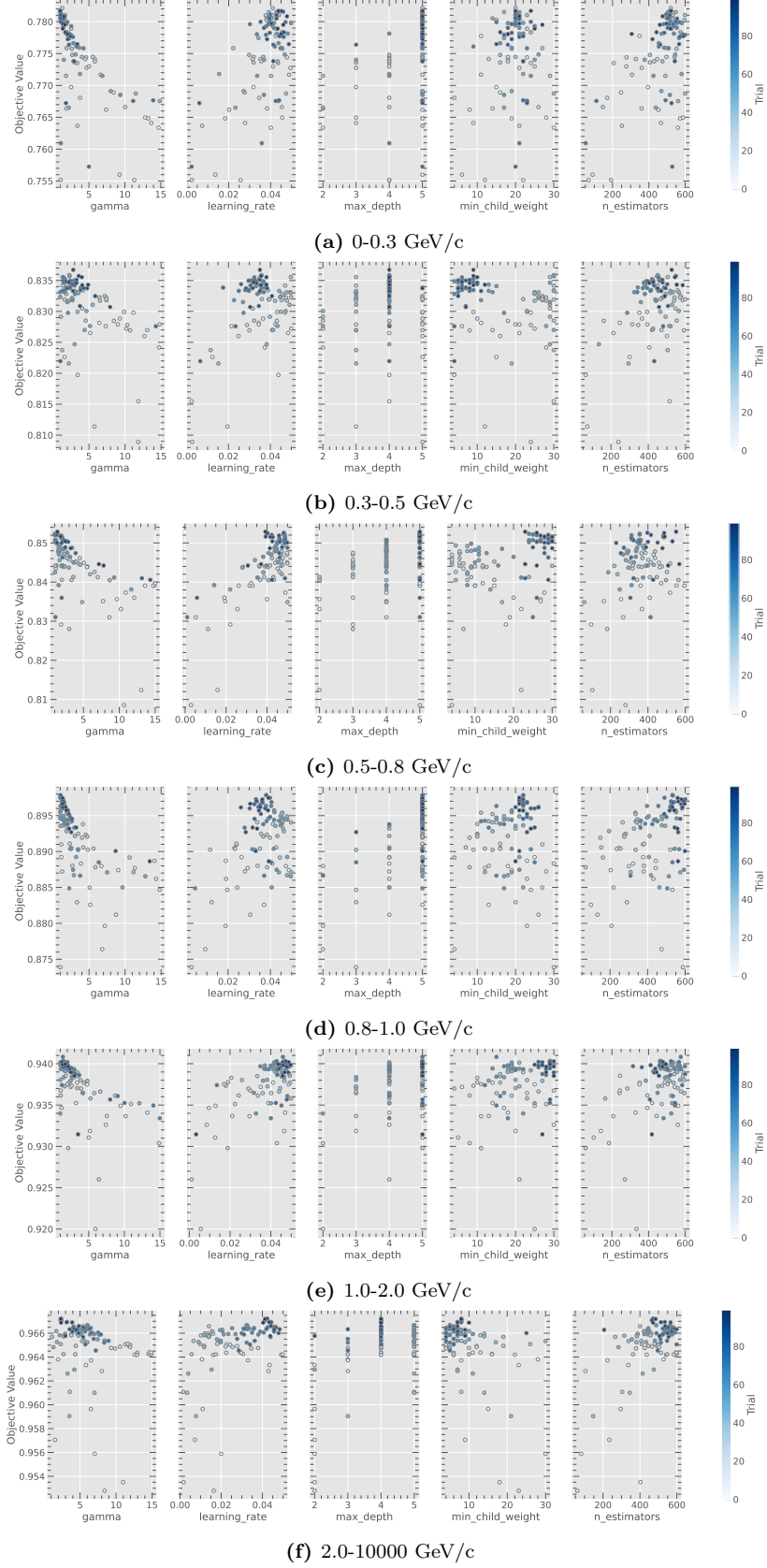


Figure 11: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 60-80 %

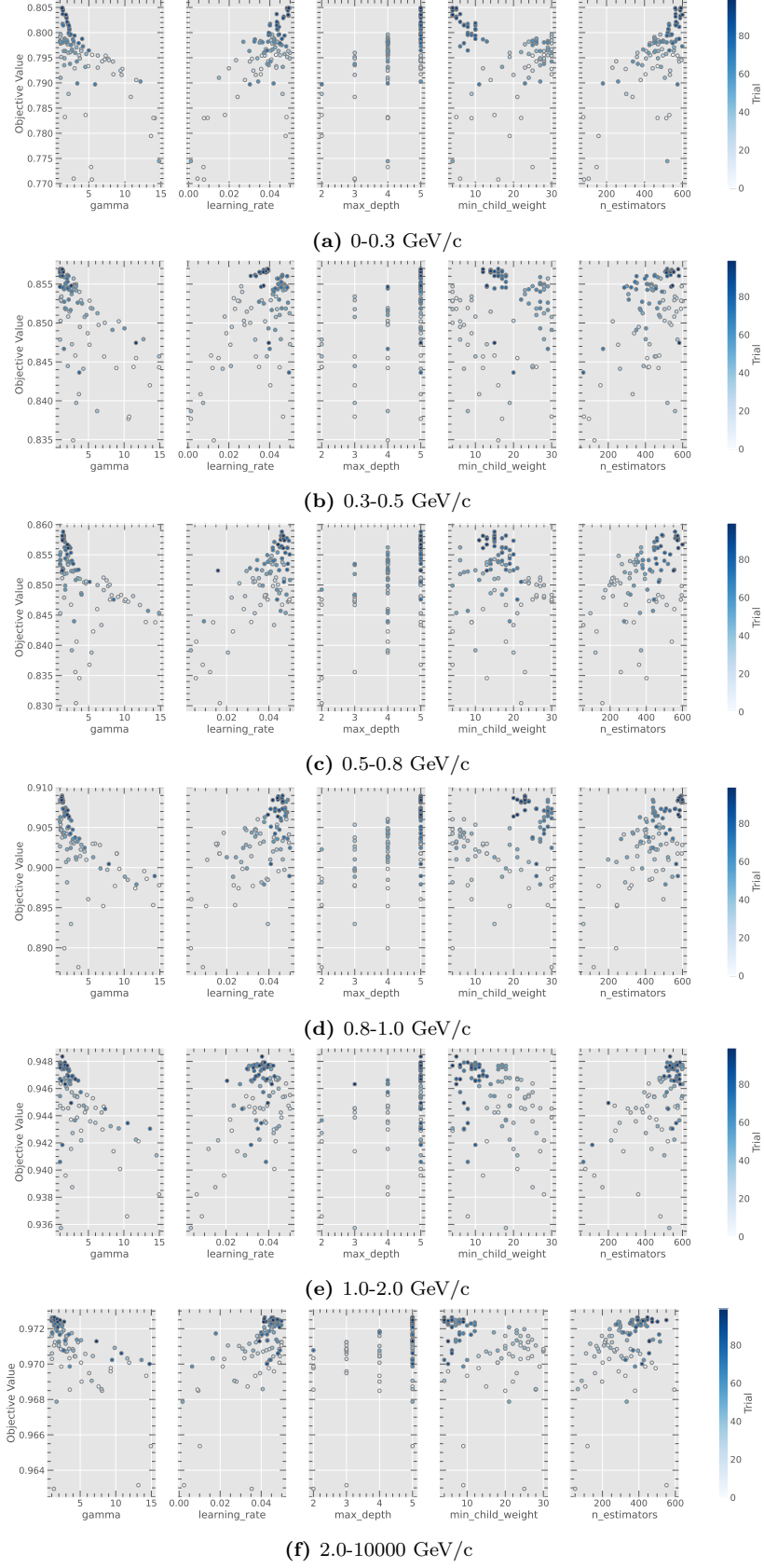


Figure 12: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 80-100 %

2.2 Colsample by tree, Lasso and Ridge regularisation, class imbalance, and subsample

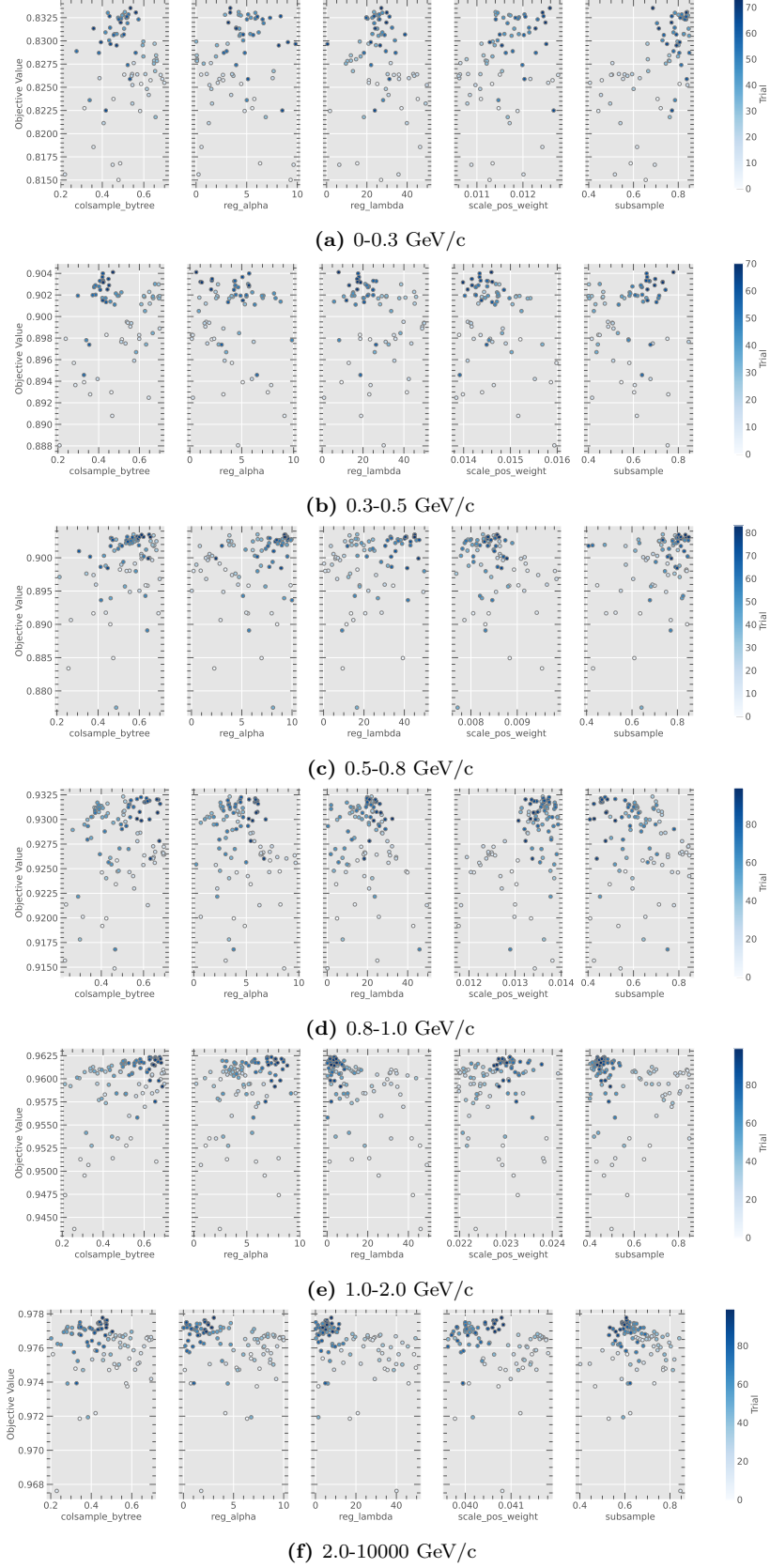


Figure 13: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 0-10 %



Figure 14: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 10-20 %

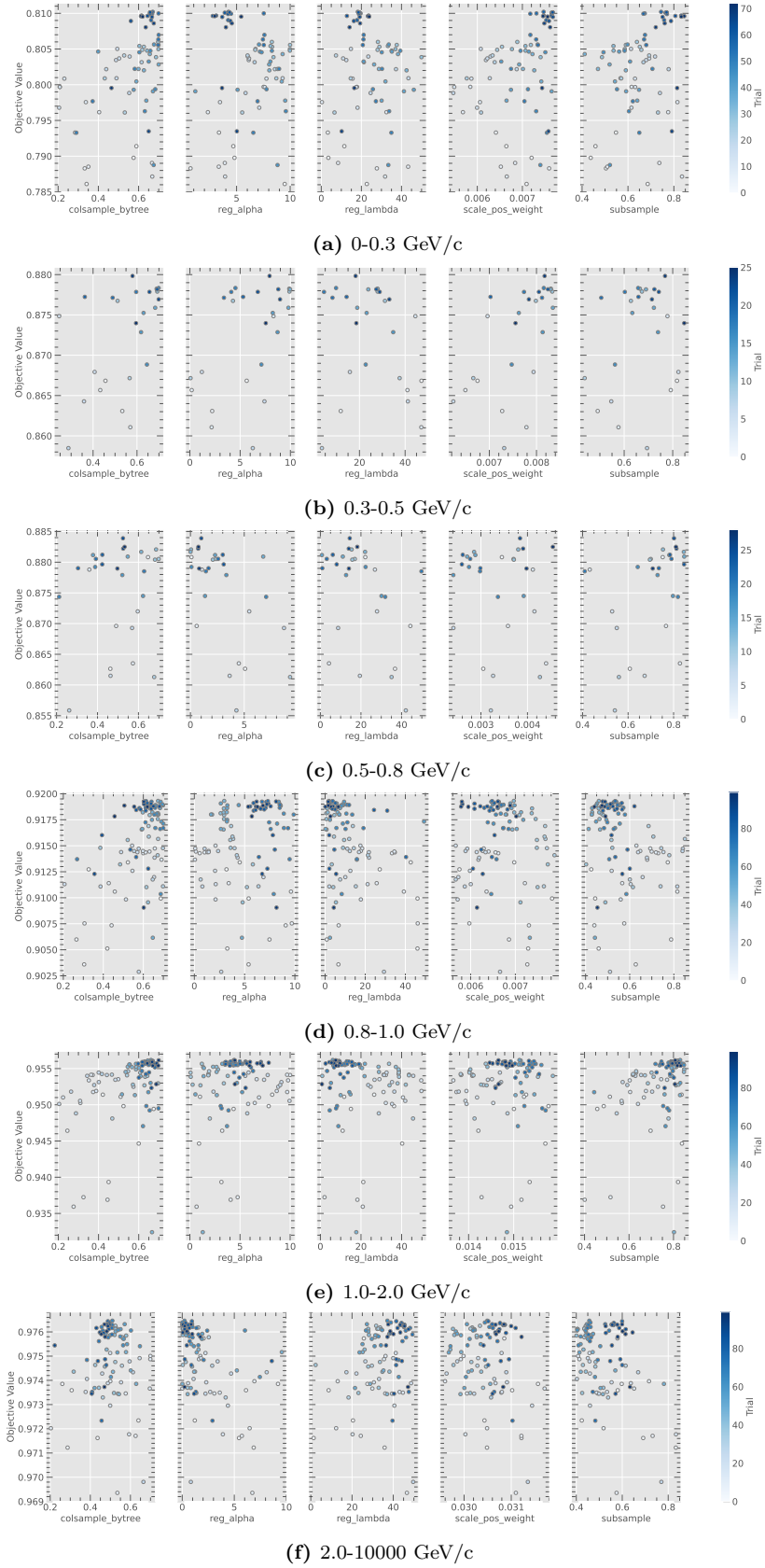


Figure 15: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 20-40 %



Figure 16: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 40-60 %

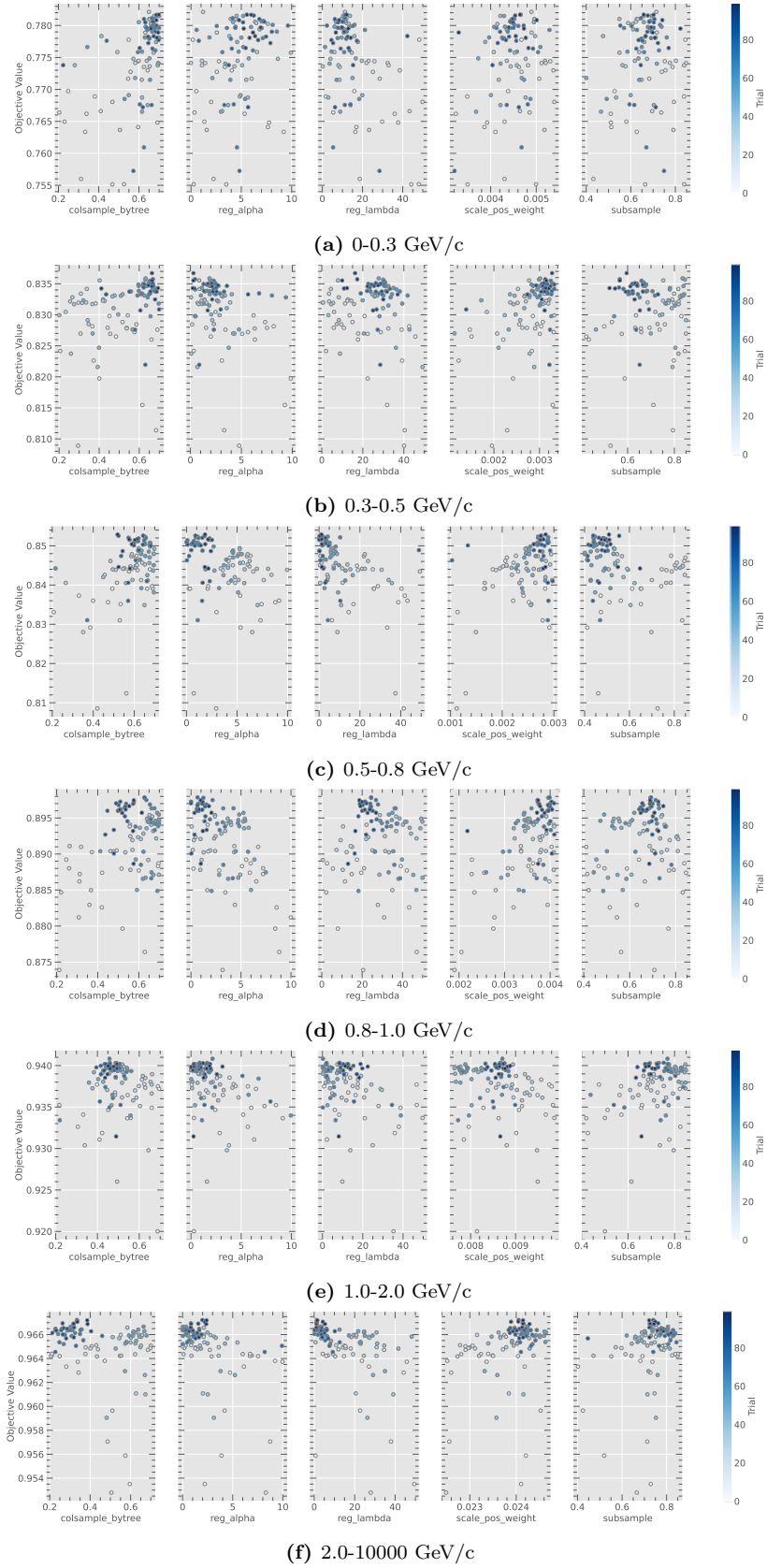


Figure 17: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 60-80 %

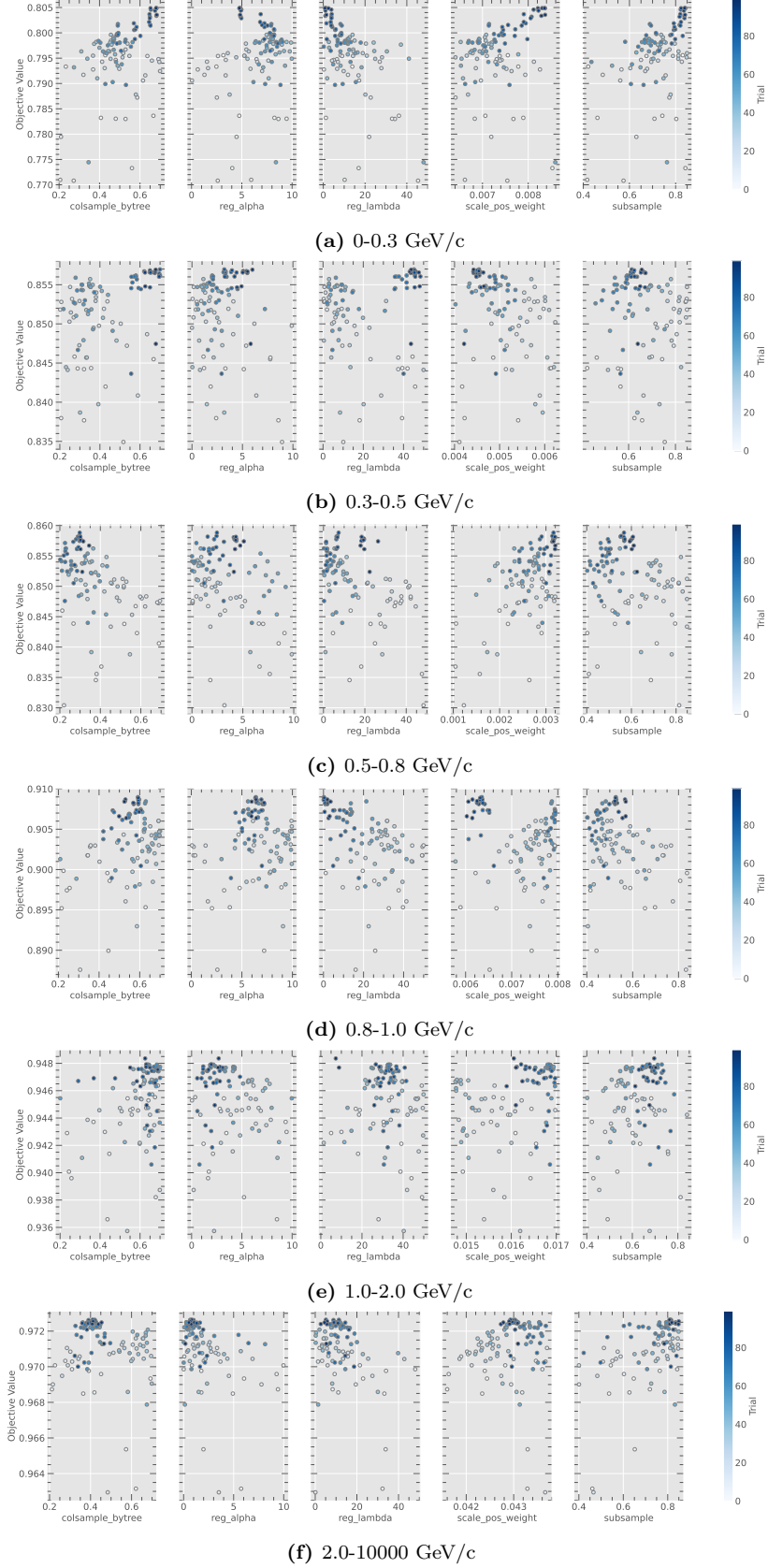
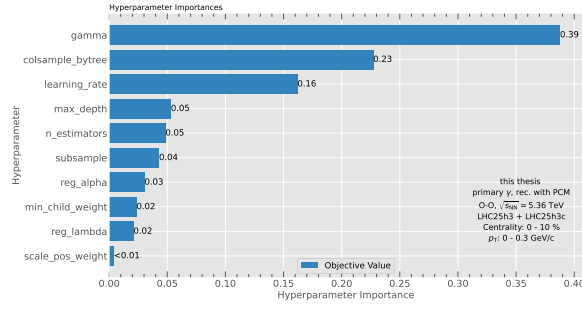
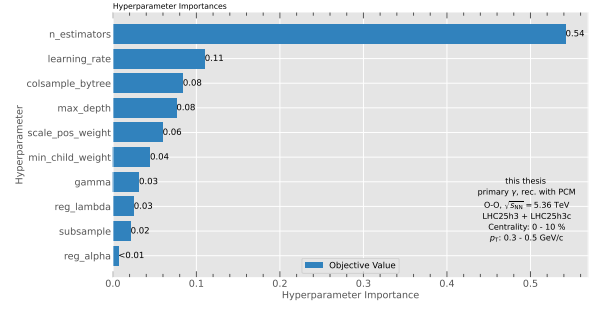


Figure 18: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 80-100 %

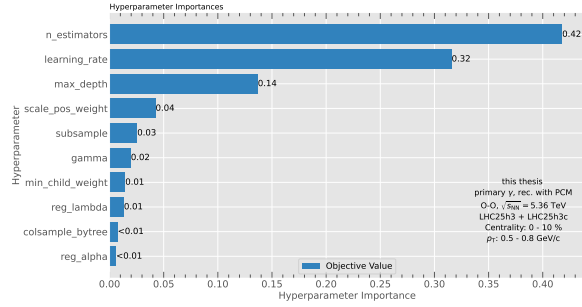
3 Hyperparameter importance



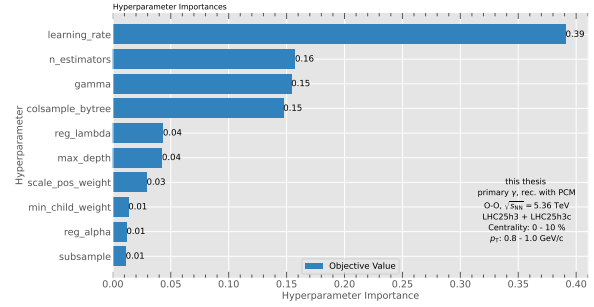
(a) 0-0.3 GeV/c



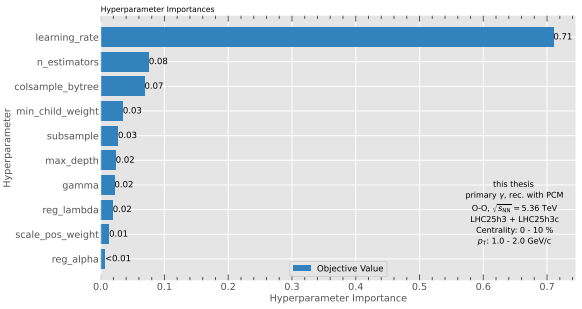
(b) 0.3-0.5 GeV/c



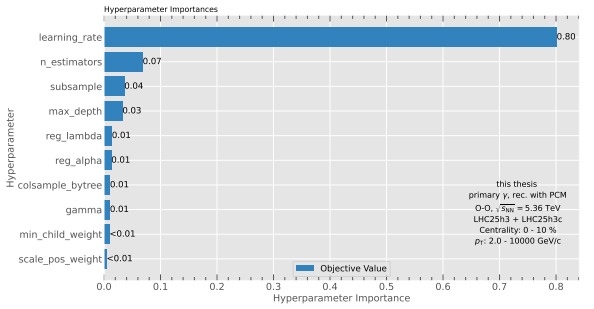
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

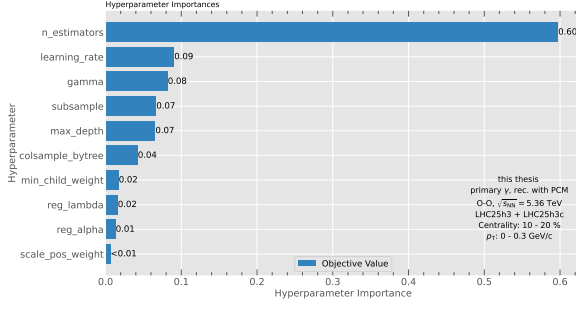


(e) 1.0-2.0 GeV/c

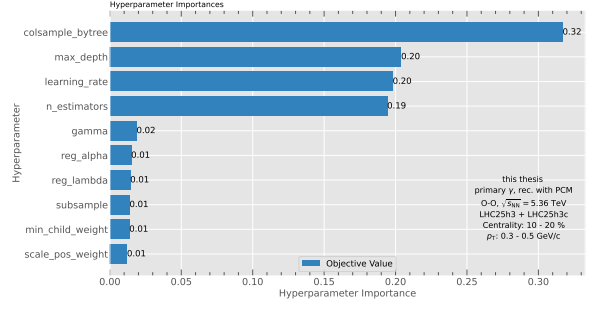


(f) 2.0-10000 GeV/c

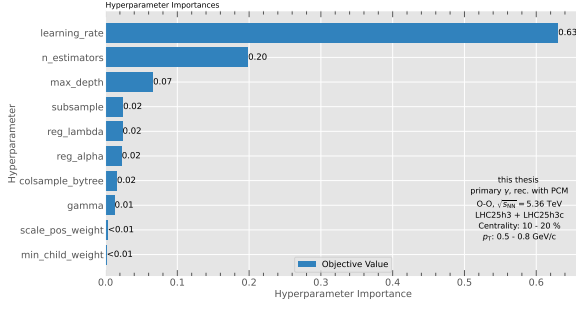
Figure 19: Hyperparameter importance for all p_T bins in the centrality range 0-10 %



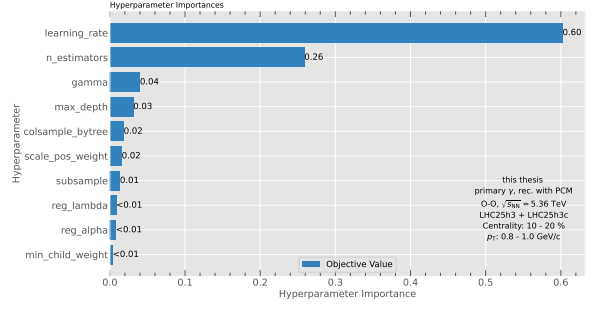
(a) 0-0.3 GeV/c



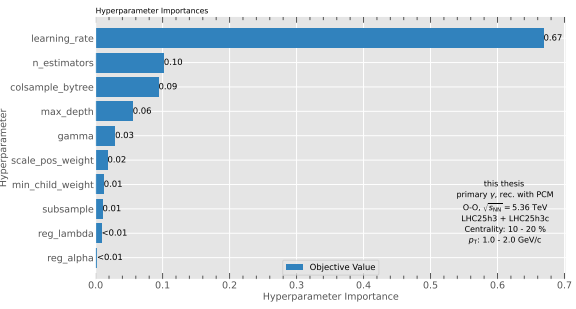
(b) 0.3-0.5 GeV/c



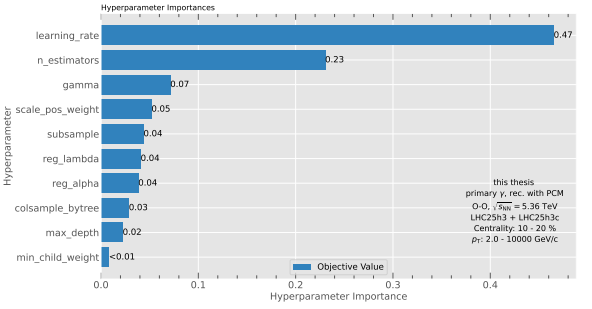
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

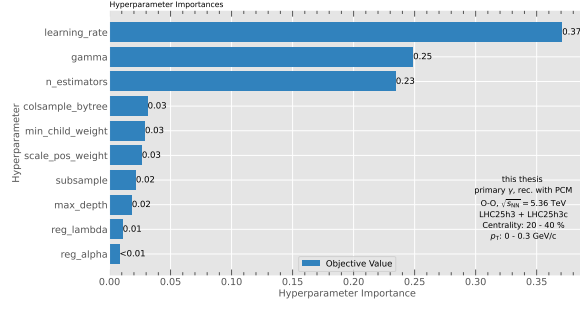


(e) 1.0-2.0 GeV/c

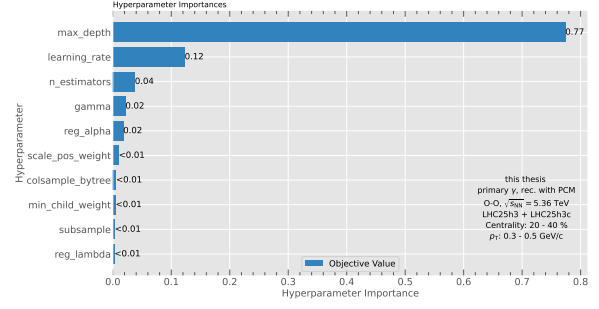


(f) 2.0-10000 GeV/c

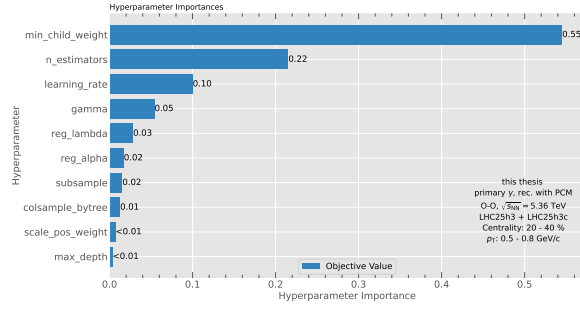
Figure 20: Hyperparameter importance for all p_T bins in the centrality range 10-20 %



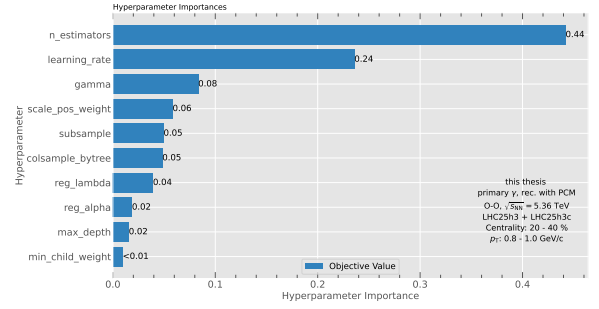
(a) 0-0.3 GeV/c



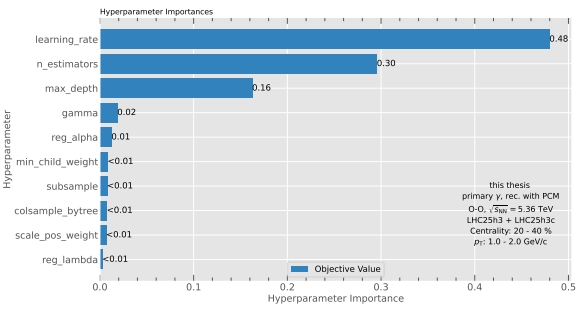
(b) 0.3-0.5 GeV/c



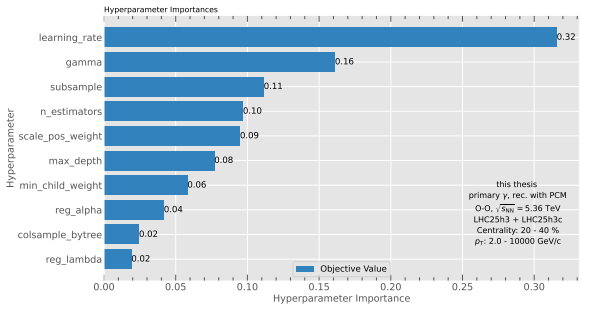
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

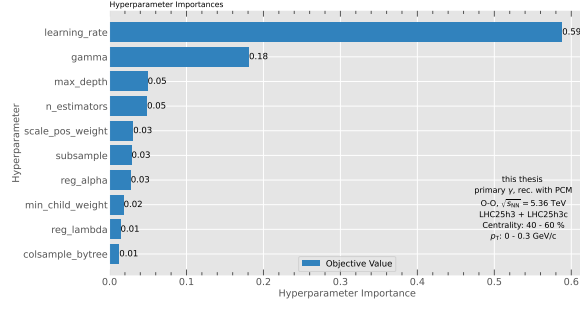


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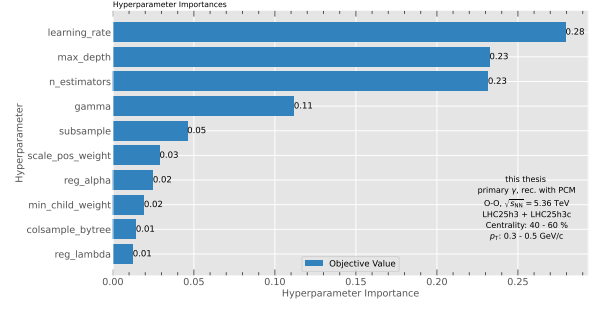


(f) 2.0-10000 GeV/c

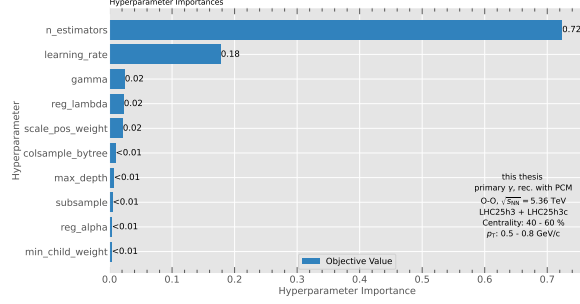
Figure 21: Hyperparameter importance for all p_T bins in the centrality range 20-40 %



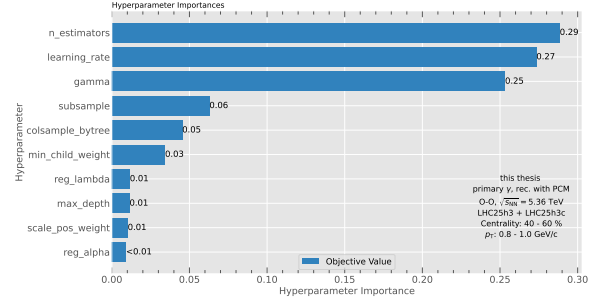
(a) 0-0.3 GeV/c



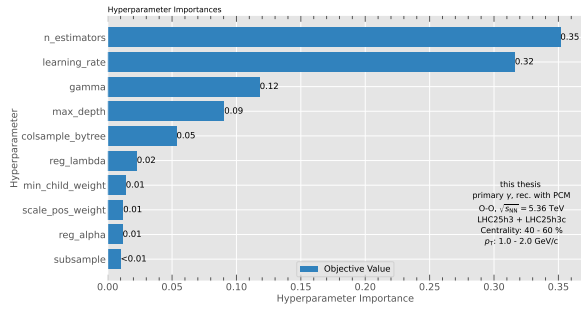
(b) 0.3-0.5 GeV/c



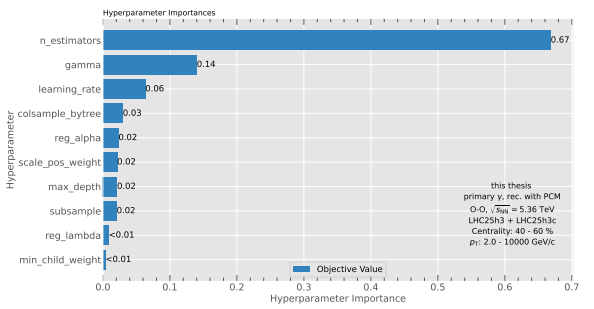
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

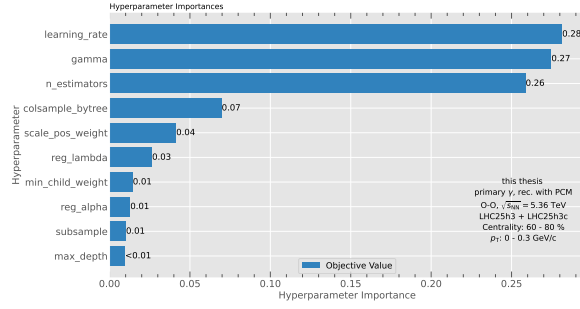


(e) 1.0-2.0 GeV/c

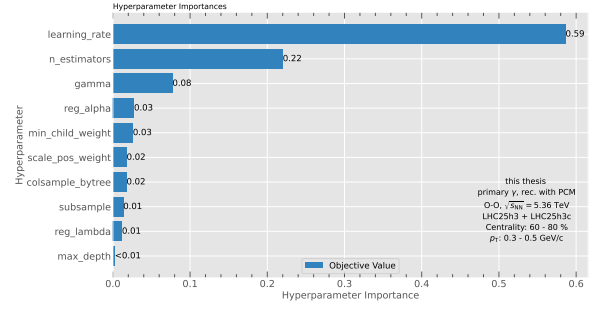


(f) 2.0-10000 GeV/c

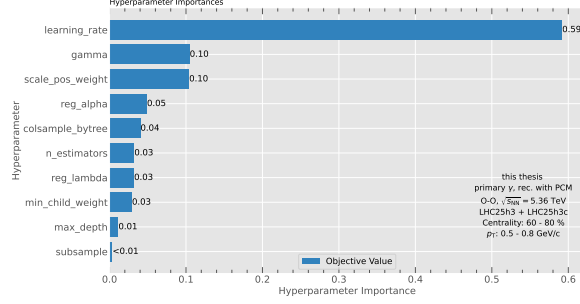
Figure 22: Hyperparameter importance for all p_T bins in the centrality range 40-60 %



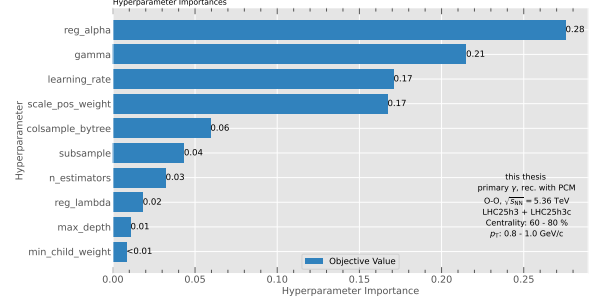
(a) 0-0.3 GeV/c



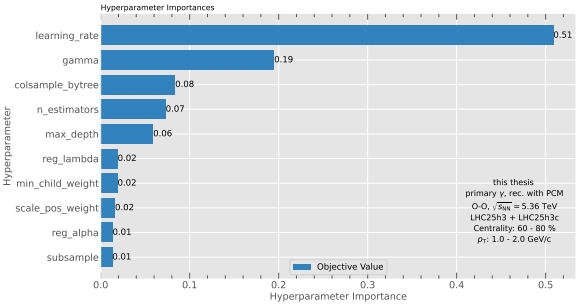
(b) 0.3-0.5 GeV/c



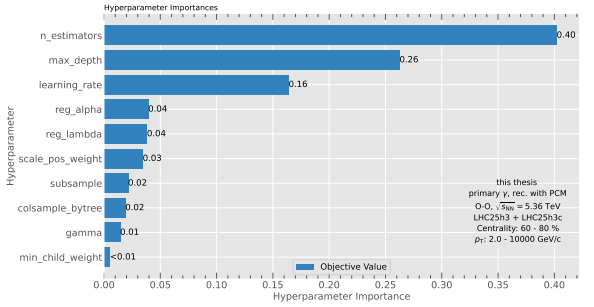
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c

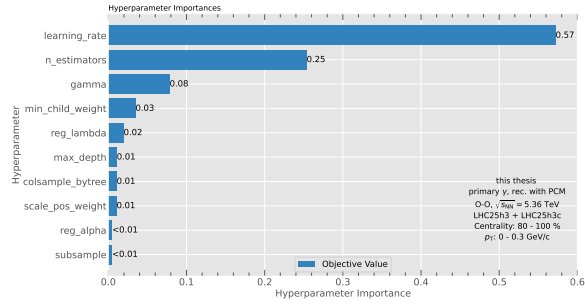


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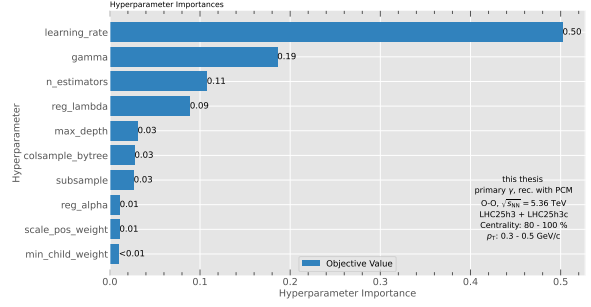


(f) 2.0-10000 GeV/c

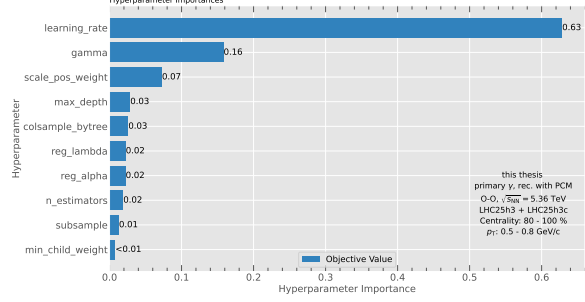
Figure 23: Hyperparameter importance for all p_T bins in the centrality range 60-80 %



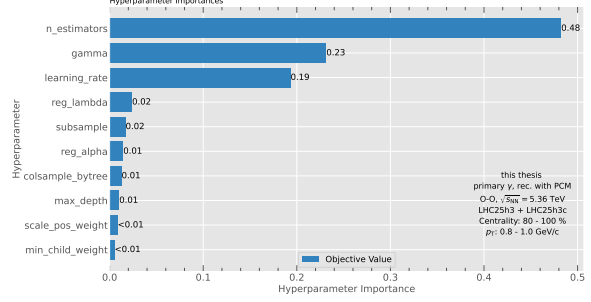
(a) 0-0.3 GeV/c



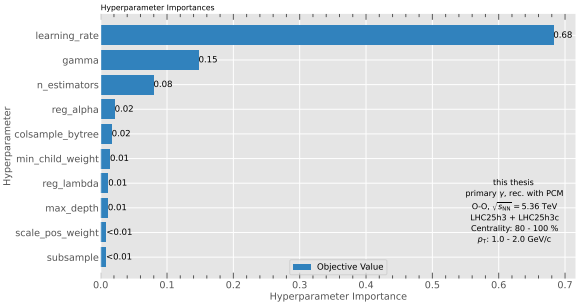
(b) 0.3-0.5 GeV/c



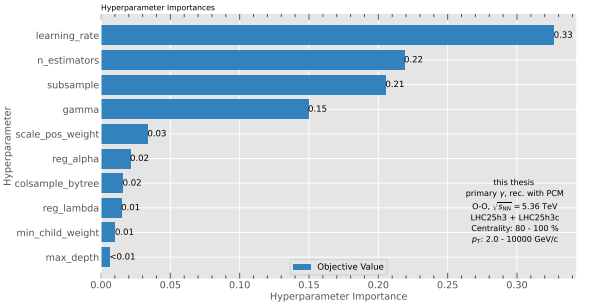
(c) 0.5-0.8 GeV/c



(d) 0.8-1.0 GeV/c



(e) 1.0-2.0 GeV/c



(f) 2.0-10000 GeV/c

Figure 24: Hyperparameter importance for all p_T bins in the centrality range 80-100 %